Merchant

Time Limit	**50** test inputs must be executed within **1.5** second(s) using C/C++, or **1.5** second(s) using Java
Memory Limit	Stack: **1** MB / Total: **256** MB
Code Length Limit	**128** KB
Evaluation	When you submit an answer, the result is shown in real time. **"Pass"** means the answer is correct for all of the test inputs. **"Fail"** means the answer is incorrect for all or part of the test inputs.

Problem

Trader Sangok Im is the leader of a merchant group that travels around cities, trading in goods and making money.

The city Sangok is about to arrive in is made up of R × C towns, arranged in an R × C grid plan (R rows × C columns). Sangok and his fellow merchants plan to stay in this city for D days before moving on to the next city.

Each day, the merchant group can do exactly one of the following:

- **Travel**
  Move the merchant group to a neighbouring town.
- **Produce**
  Make goods to sell. Goods can be produced several times in each town, but each production uses up materials, so the **next production in that town is halved** (rounded down to an integer).
- **Trade**
  Sell goods. Goods can be traded several times, but each sale lowers demand, so the **selling price in that town drops by 1 won**.

Sangok has researched the market and estimated how much one piece of goods is worth in each town, and how many pieces of goods can be produced in each town using the town's production capacity and materials.

Now that the market research is complete, Sangok wants to calculate how much profit he can make in this city.

5/1	7/2	8/2	2/2
7/2	22/5	6/1	7/13
5/5	5/5	3/1	10/3
3/2	1/5	9/4	8/3
32/10	2/5	5/1	5/3

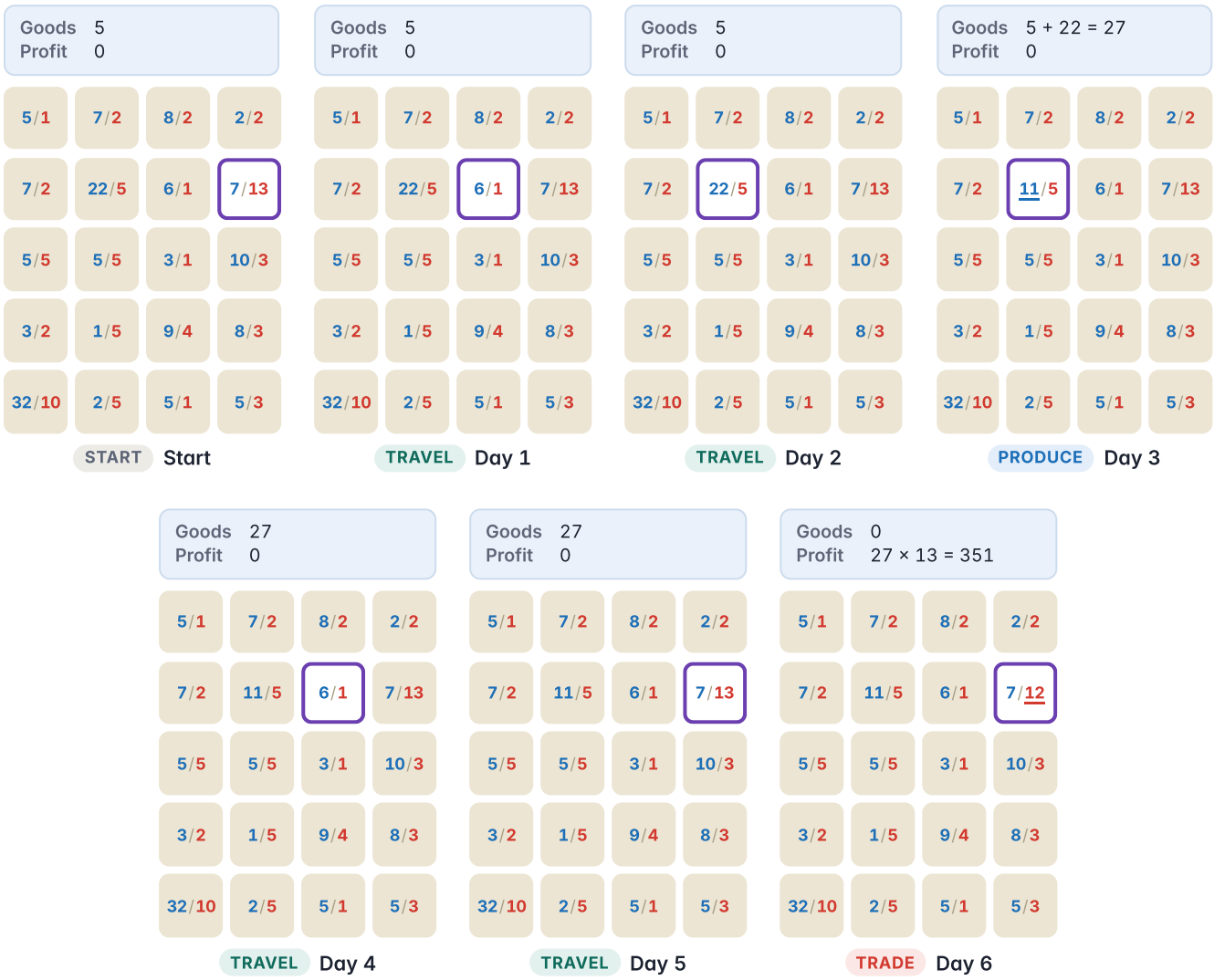
[Fig. 1]

[Fig. 1] shows a city made up of 5 × 4 towns.

- The **blue** number in a box is the quantity of goods that can be produced in that town.
- The **red** number is the current price (won) per piece of goods at which the merchant group plans to sell.
- In the examples below, the outlined town is where the merchant group currently is; an underlined number has just changed.

Example 1

The merchant group plans to stay in this city for **6 days**, starting at town **(2, 4)**. Assume the merchant group currently owns **5** pieces of goods.



> Travelling (2, 4) → (2, 3) → (2, 2) takes two days. The group then produces goods at town (2, 2), travels (2, 2) → (2, 3) → (2, 4) again and sells the goods at town (2, 4).
>
> Doing all of this, the merchants make a profit of **351 won** by selling a total of 27 pieces of goods at 13 won/piece.

Example 2

Now let's change some of the conditions. The merchant group plans to stay in the same city for **4 days**, starting at town **(5, 1)**. Assume the merchant group currently owns **5** pieces of goods.



> By producing goods three times and then trading at town (5, 1), the merchants make a profit of **610 won** by selling a total of 61 pieces of goods at 10 won/piece.

Task

Given the city in which the merchants plan to stay, the duration of the stay, the number of goods each town can produce and the current price of goods in each town, write a program that calculates the **maximum profit** Sangok's merchant group can make.

Constraints

1. R and C give the size of the city (R rows, C columns). R and C are natural numbers (1 ≤ R, C ≤ 20).
2. D, the duration of the merchants' stay in the city, is a natural number (1 ≤ D ≤ 9).
3. I, the quantity of goods initially owned by the merchant group, is a natural number (1 ≤ I ≤ 100,000).
4. The quantity of goods each town can produce is a natural number (1 ≤ quantity ≤ 100,000).
5. The price of one piece of goods in each town is a natural number (1 ≤ price ≤ 100).

Input

The first line contains **T**, the number of test cases, followed by T test cases.

The first line of each test case contains six integers separated by spaces: **R** (number of rows), **C** (number of columns), **SR** (starting row), **SC** (starting column), **D** (number of days) and **I** (initial quantity of goods).

The next **R lines** each contain C integers: the quantity of goods each town can produce. The following **R lines** each contain C integers: the price of one piece of goods in each town. Values on a line are separated by spaces.

`T`	number of test cases (once, at the top)
`R C SR SC D I`	first line of each test case
$P_{1,1}\ P_{1,2}\ \cdots\ P_{1,C}$ $\vdots$ $P_{R,1}\ P_{R,2}\ \cdots\ P_{R,C}$	R lines — quantity of goods each town can produce
$V_{1,1}\ V_{1,2}\ \cdots\ V_{1,C}$ $\vdots$ $V_{R,1}\ V_{R,2}\ \cdots\ V_{R,C}$	R lines — price of one piece of goods in each town

Output

For each test case, print `#x` (x is the test case number, starting from 1), a space, then the maximum profit Sangok's merchant group can make in this city. Print one line per test case.

Sample Input / Output

INPUT

```
2
5 4 2 4 6 5
5 7 8 2
7 22 6 7
5 5 3 10
3 1 9 8
32 2 5 5
1 2 2 2
2 5 1 13
5 5 1 3
2 5 4 3
10 5 1 3
5 4 5 1 4 5
5 7 8 2
7 22 6 7
5 5 3 10
3 1 9 8
32 2 5 5
1 2 2 2
2 5 1 13
5 5 1 3
2 5 4 3
10 5 1 3
```

OUTPUT

```
#1 351
#2 610
```

OUTPUT OF SAMPLE_INPUT.TXT

```
#1 351
#2 610
#3 1424
#4 34840
#5 0
#6 199950
#7 100520
#8 231252
#9 69160
#10 231600
#11 134292
#12 541985
#13 138524
#14 0
#15 149361
```

```
#16 1483774
#17 13356
#18 0
#19 15465170
#20 2742846
#21 916760
#22 138840
#23 713320
#24 130440
#25 343030
#26 10289664
#27 19112544
#28 40538190
#29 0
#30 39902880
```